

SOLAX POWER | BVB 09 PARTNER

AELIO+HR140

49.9kW/50kW/60kW/61kW
100.3kWh~401.4kWh

SolaX Split C&I Energy Storage Solution

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X3-AELIO



** Feature to be upgraded in the future*

Flexible Configuration

- Customizable Battery Setup: Batteries can be flexibly configured according to customer site requirements, supporting both indoor and outdoor installations, as well as rack-mounted and stacked installations
- Parallel Operation: Supports up to 10 parallel systems, meeting power requirements from 49.9 kW to 600 kW
- Battery Options: Two battery choices available, catering to single system capacities ranging from 100.3 kWh to 401.4 kWh
- PV Oversizing: Supports up to 200% PV oversizing

Safe and Reliable

- Three-Phase Imbalance: Inverter supports 100% three-phase imbalance
- Long-Term Overload: Inverter supports 110% long-term overload
- Seamless Switching: Inverter supports 10ms on-grid/off-grid switching, ensuring uninterrupted power supply for onsite loads
- Off-Grid Overload: Inverter supports 150% off-grid overloading

Intelligent and Convenient

- Remote Monitoring: Supports web/app remote data viewing and remote OTA updates
- Versatile Operating Modes: Supports various operating modes to meet different application scenarios such as self-consumption, peak shaving, and demand management
- VPP Integration: Supports VPP applications through 2030.5, OpenADR and API*
- Generator Control: Work seamlessly with generator to save fuel and back up

	X3-AELIO-49.9K	X3-AELIO-49.9K-P	X3-AELIO-50K	X3-AELIO-60K	X3-AELIO-61K
PV INPUT					
Max. recommended PV array power	100 kWp	120 kWp	100 kWp	120 kWp	
Max. PV input voltage①	1000 V				
Nominal PV input voltage	650 V				
MPPT voltage range②	160 ~ 950 V				
Start-up voltage	180 V				
No. of MPP trackers / Strings per MPP tracker	5 / 2	6 / 2	5 / 2	6 / 2	
Max. input current per MPPT	40 A				
Max. input short circuit current per MPPT	50 A				
AC INPUT & OUTPUT (ON-GRID)					
Rated output power	49.9 kW	49.9 kW	50 kW	60 kW	61 kW
Rated output current	75.7 A @ 220 V 72.4 A @ 230 V 69.4 A @ 240 V	75.7 A @ 220 V 72.4 A @ 230 V 69.4 A @ 240 V	75.8 A @ 220 V 72.5 A @ 230 V 69.5 A @ 240 V	91.0 A @ 220 V 87.0 A @ 230 V 83.4 A @ 240 V	92.5 A @ 220 V 88.5 A @ 230 V 84.8 A @ 240 V
Max. output apparent power	49.9 kVA	49.9 kVA	50.0 kVA	60.0 kVA	61.0 kVA
Max. output continuous current	75.7 A @ 220 V 72.4 A @ 230 V 69.4 A @ 240 V	75.7 A @ 220 V 72.4 A @ 230 V 69.4 A @ 240 V	83.4 A @ 220 V 79.8 A @ 230 V 76.4 A @ 240 V	100.0 A @ 220 V 95.7 A @ 230 V 91.7 A @ 240 V	100.0 A @ 220 V 95.7 A @ 230 V 91.7 A @ 240 V
Nominal AC voltage	3 / N / PE, 220 / 380 V 3 / N / PE, 230 / 400 V 3 / N / PE, 240 / 415 V				
Nominal AC frequency	50 Hz / 60 Hz				
AC frequency range③	50 ± 5 Hz / 60 ± 5 Hz				
Adjustable Power Factor range	~ 1 (0.8 lagging to 0.8 leading)				
THDi (rated power)	< 3%				

	X3-AELIO-49.9K	X3-AELIO-49.9K-P	X3-AELIO-50K	X3-AELIO-60K	X3-AELIO-61K
Battery type	Lithium				
Battery voltage range	160 ~ 820 V				
Max. charge / discharge current	160 A (80 A × 2)				
Rated EPS output voltage, frequency	3 / N / PE, 220 / 380 V, 50 Hz / 60 Hz 3 / N / PE, 230 / 400 V, 50 Hz / 60 Hz 3 / N / PE, 240 / 415 V, 50 Hz / 60 Hz				
Rated EPS output power	49.9 kVA	49.9 kVA	50.0 kVA	60.0 kVA	61.0 kVA
Peak EPS output power	75 kVA, 10s	75 kVA, 10s	75 kVA, 10s	90 kVA, 10s	90 kVA, 10s
Switchover time	< 10 ms				
Max. efficiency	98.0%				
European efficiency	97.2%				
Ingress protection	IP66				
Operating ambient temperature range④	-35 ~ 60°C				
Max. operating altitude	3000 m				
Relative humidity	0 ~ 100% RH (condensing)				
Overvoltage Category	Mains: III, Battery: II, PV: II				
Dimensions (W × H × D)	820 × 670 × 257 mm				
Net weight	< 100 kg	< 105 kg	< 100 kg	< 105 kg	< 105 kg
Cooling concept	Smart air cooling				
Communication interfaces	RS485-Meter, RS485-Monitor, RS485-Parallel(daisy-chain), CAN-BMS, CAN-Parallel(daisy-chain), USB, DI×2, DO×1, RCR(DI×4), DRM				
Topology	Non-isolated				
Certificates and approvals	CE, VDE4105, G99, AS4777, EN50549, CEI 0-21, IEC61727, PEA/MEA, NRS-097-2-1, RD1699, TOR				
	Protection				
Over / under voltage protection	Yes				
DC isolation protection	Yes				
DC reverse-polarity protection	Yes				
Grid monitoring	Yes				
DC injection monitoring	Yes				
Back feed current monitoring	Yes				
Residual current detection	Yes				
Over temperature protection	Yes				
Active anti-islanding method	Frequency shift				
Surge protection (DC / AC)	DC: Type II, AC: Type II				
Arc-fault circuit interrupter (AFCI)	Optional				
AC auxiliary power supply (APS)	Built-in				

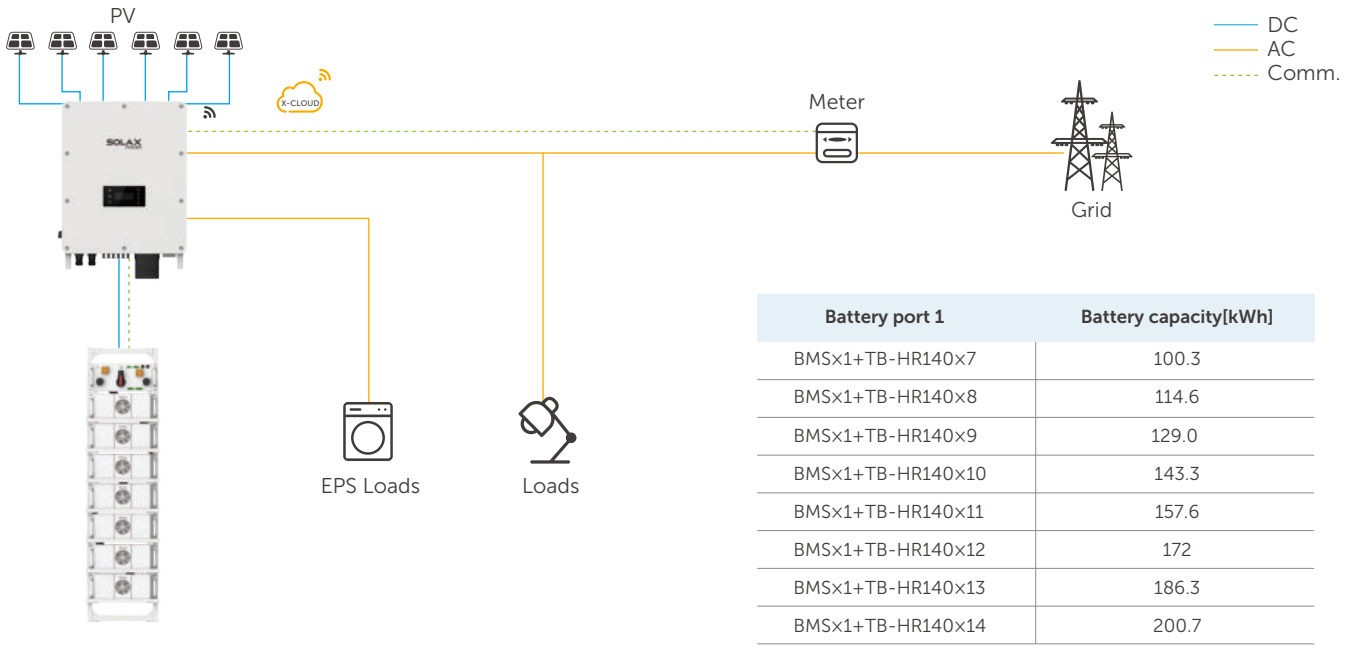
① The maximum input voltage is the upper limit of the DC voltage. Any higher input DC voltage would probably damage inverter
② Input voltage exceeding the MPPT voltage range may triggers inverter protection
③ The AC frequency range may vary from different country codes
④ Derating above +45°C

TSYS-HR140

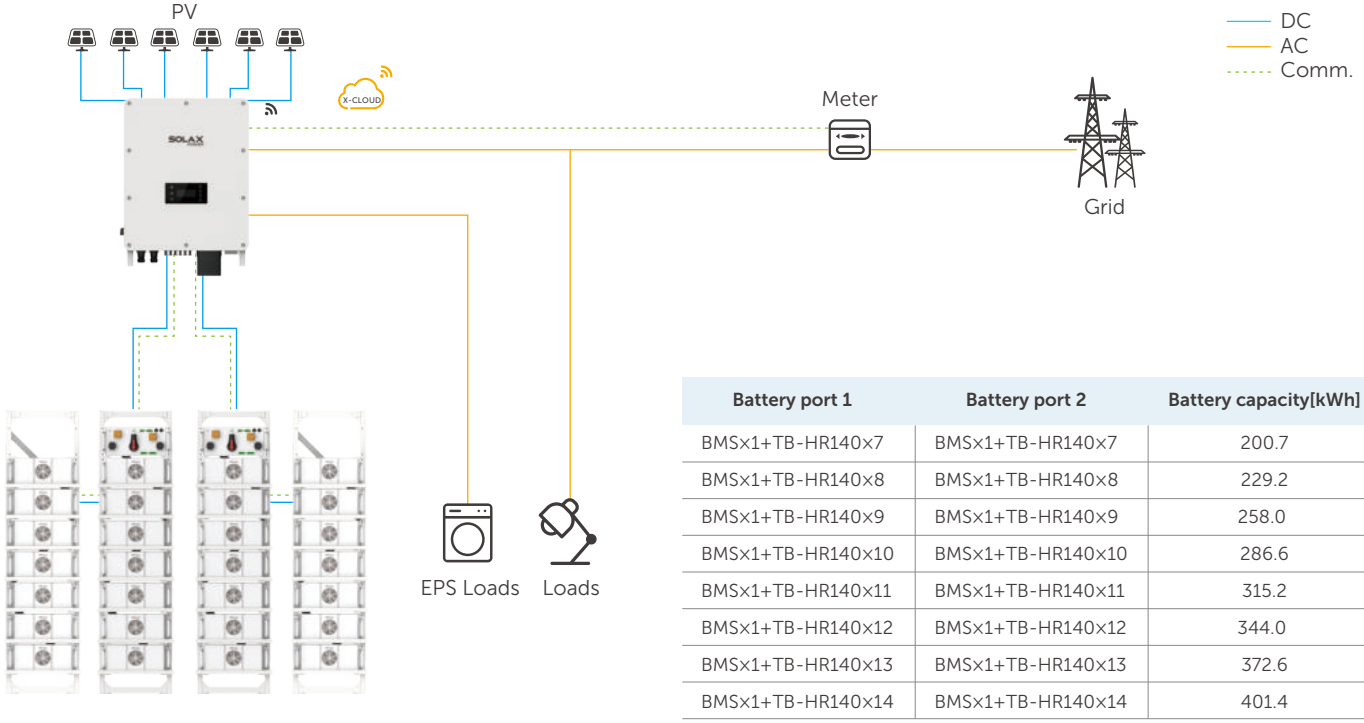


	T-HR1003	T-HR1146	T-HR1290	T-HR1433	T-HR1576	T-HR1720	T-HR1863	T-HR2007
SYSTEM SPECIFICATION								
Module quantity	7	8	9	10	11	12	13	14
Nominal capacity [kWh]	100.3 kWh	114.6 kWh	129.0 kWh	143.3 kWh	157.6 kWh	172.0 kWh	186.3 kWh	200.7 kWh
Usable energy (90%DOD)①	90.3 kWh	103.1 kWh	116.1 kWh	129.0 kWh	141.8 kWh	154.8 kWh	167.7 kWh	180.6 kWh
Nominal voltage	358.4 V	409.6 V	460.8 V	512 V	563.2 V	614.4 V	665.6 V	716.8 V
Operating voltage range	291 ~ 408 V	332 ~ 467 V	374 ~ 526 V	416 ~ 584 V	457 ~ 642 V	499 ~ 700 V	540 ~ 759 V	582.4 ~ 817.6 V
Nominal operation current③	140							
Maximum operation current②③	140							
Norminal power③	50 kW	57 kW	65 kW	72 kW	79 kW	86 kW	93 kW	100 kW
Depth of discharge	90%							
Model	TB-HR140							
Module energy	14.3 kWh							
Rated module voltage	51.2 V							
Module capacity	280 Ah							
Battery type	LiFePO4							
Dimension (W x H x D)	455 × 228 × 731 mm							
BMS								
Model	TBMS-R15							
Dimension (W x H x D)	461 × 228 × 778 mm							
Weight	31.3 kg							
GENERAL SPECIFICATION								
Cycel Life (90% DOD)	8000							
Charge / Discharge Temperature Range (Without Heating)	0°C to 53°C (Charge) -20°C to 53°C (Discharge)							
Communication interfaces	CAN							
Installation type	Rack Mounting							
Relative humidity	0 ~ 95%(RH)							
Altitude	＜3000 m							
Environment	Indoor							
Protection degree	IP20							
Safety	EN 62477, IEC 62619, IEC 61000-6-1/2/3/4							
Transport testing requirement	UN38.3							

BATTERY MODULE COMBINATION TABLE



Battery port 1	Battery capacity[kWh]
BMSx1+TB-HR140x7	100.3
BMSx1+TB-HR140x8	114.6
BMSx1+TB-HR140x9	129.0
BMSx1+TB-HR140x10	143.3
BMSx1+TB-HR140x11	157.6
BMSx1+TB-HR140x12	172
BMSx1+TB-HR140x13	186.3
BMSx1+TB-HR140x14	200.7



Battery port 1	Battery port 2	Battery capacity[kWh]
BMSx1+TB-HR140x7	BMSx1+TB-HR140x7	200.7
BMSx1+TB-HR140x8	BMSx1+TB-HR140x8	229.2
BMSx1+TB-HR140x9	BMSx1+TB-HR140x9	258.0
BMSx1+TB-HR140x10	BMSx1+TB-HR140x10	286.6
BMSx1+TB-HR140x11	BMSx1+TB-HR140x11	315.2
BMSx1+TB-HR140x12	BMSx1+TB-HR140x12	344.0
BMSx1+TB-HR140x13	BMSx1+TB-HR140x13	372.6
BMSx1+TB-HR140x14	BMSx1+TB-HR140x14	401.4

① Test conditions: 90% DOD, 0.2C charger & discharger @ 25°C
② Max. charge / discharge current may be variant with different inverter models
③ Nominal / Maximum operation current and nominal / maximum power derating will occur related to temperature or SOC



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